

Chosen Model for Bike Rentals Predictor

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1. Introduction

For this project, I selected **Linear Regression** as my first machine learning model. It is simple, clear, and easy to understand, which makes it a good starting point. The goal of my project is to predict the **number of daily bike rentals** using weather and time-related features. Linear Regression is a **supervised regression model** that fits this goal perfectly because it predicts numerical values.

2. Why Linear Regression

I chose Linear Regression because it is:

- **Easy to interpret** and shows how each factor affects the number of rentals.
 - **Beginner-friendly** and useful for understanding how models learn from data.
 - **Lightweight and fast**, so it can train quickly even on large datasets.
 - **A good baseline model** that helps compare future, more complex models like Random Forest or Gradient Boosting.
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3. How the Model Works

Linear Regression tries to find a straight line that best fits the data. The formula looks like this:

$$\hat{y} = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$$

where:

- x_i are the input features (temperature, humidity, weekday, etc.)
- w_i are the weights that show how important each feature is
- b is the bias term
- y is the predicted number of rentals

The model learns by minimizing the difference between the predicted and actual values using a process called **Gradient Descent**. This process helps the model reduce errors step by step.

4. Evaluation Plan

To check how accurate the model is, I will use:

- **RMSE (Root Mean Squared Error):** Tells how large the prediction errors are on average. A lower RMSE means the model's predictions are closer to the real values.
- **MAE (Mean Absolute Error):** Measures the average distance between predicted and actual values, showing how much the model is off on a typical day.
- **R^2 (Coefficient of Determination):** Explains how well the model captures the pattern in the data. A higher R^2 means the model understands more of the changes in bike rentals.

Lower RMSE and MAE and **higher** R^2 mean better performance.

5. Modeling and Improvement Steps

1. Model Training

I will train the Linear Regression model using my cleaned and prepared dataset. The data will be divided into two parts: one for training and one for testing. During training, the model will learn the relationships between weather and time features and the total number of daily rentals.

2. Model Testing

After training, I will test the model using data it has never seen before. This step shows how well the model can predict real-world situations. The predictions will then be compared with the actual number of rentals to calculate the error.

3. Evaluation and Interpretation

Once the model gives predictions, I will evaluate how accurate they are using RMSE, MAE, and R^2 scores. Then I will look at the model's coefficients to understand which factors have the strongest positive or negative influence on bike rentals. For example, if temperature has a high positive weight, it means warmer days usually lead to more rentals.

4. Iteration and Next Steps

If the model's accuracy is not good enough, I will first improve it by adjusting its features, checking data consistency, or tuning parameters like normalization. Only if these improvements do not work, I will consider testing another model.

6. Expected Outcome

Using Linear Regression, I expect to:

- Identify which weather and time factors affect bike rentals the most.
- Build a clear and reliable baseline model for predicting daily rentals.
- Use the results to make improvements and support smarter decisions for bike-sharing management.